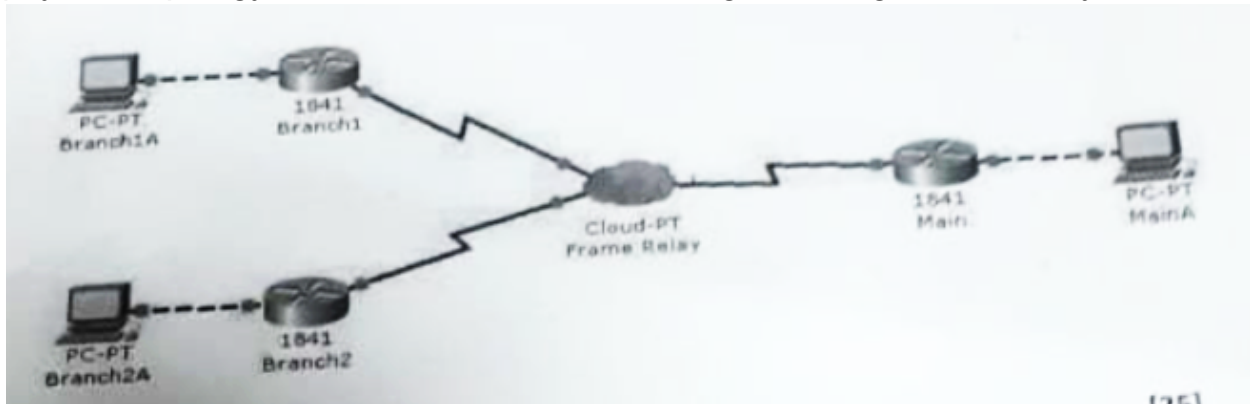


Question #1:

Demonstrate the use of variations in SHOW commands that will give a clear picture of how the following network is communicating and transferring data. The physical topology of the network has been designed using Frame Relay.

**Answer:****Use of SHOW Commands**

In Frame Relay networks, **show** commands are used to monitor, verify, and troubleshoot configuration and connectivity. The most important commands are:

1. **show running-config:** Displays current configuration, including Frame Relay encapsulation and DLCI mappings.
2. **show ip interface brief:** Shows IP addresses and interface status (up/down) in a summary form.
3. **show interfaces serial 0/0:** Provides detailed information such as encapsulation, bandwidth, errors, and LMI type.
4. **show frame-relay lmi:** Verifies LMI keepalive messages between router and Frame Relay switch.
5. **show frame-relay pvc:** Checks the PVC (Permanent Virtual Circuit) status (Active, Inactive, Deleted).
6. **show frame-relay map:** Displays DLCI-to-IP mapping used for communication.
7. **show ip route:** Verifies the routing table to ensure remote networks are reachable.
8. **show ip interface:** see if IP is up/down, ACLs, etc.
9. **show controllers serial0/0:** verify clocking (DTE/DCE) and physical layer details when trouble.

Steps in Cisco Packet Tracer

1. **Create Topology**
 - Place 3 routers (Main, Branch1, Branch2), 3 PCs, and a Frame Relay cloud.
2. **Connections**
 - Connect router serial interfaces to the Frame Relay cloud and configure DLCIs inside the cloud properties.
3. **Router Configuration**
 - Configure serial sub-interfaces with Frame Relay encapsulation and assign DLCIs.

- Configure LAN IPs on GigabitEthernet interfaces and assign IPs to PCs with default gateways.
4. **Routing Setup**
 - Add static routes on the Main router for Branch networks.
 5. **Testing**
 - Ping from Branch1 PC to Branch2 PC. If successful, Frame Relay communication is working correctly.

Verification

- **show ip int brief** → All interfaces should be **up/up**.
- **show frame-relay pvc** → PVCs should show **Active**.
- **show frame-relay map** → DLCI and IP mappings must be correct.
- **show ip route:**
Interpretation: see connected subinterfaces and static/learned routes; ensures traffic will be forwarded onto the correct serial subinterface.
- **show controllers serial 0/0**
Use if: you see interface down or lots of errors; checks DCE/DTE, clocking.
- **Ping/Traceroute** → Verify end-to-end connectivity between PCs.

Question #2:

Use Wireshark to capture the HTTP GET and HTTP response messages when accessing

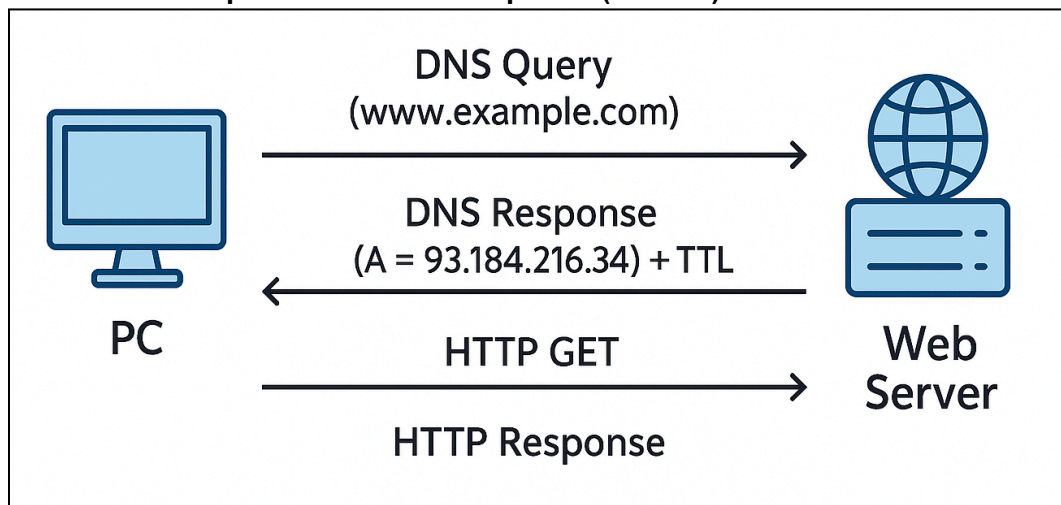
www.example.com. Identify:

- a) The HTTP request and response method and response status code & type of content returned.
- b) Open browser and clear your DNS cache and explain DNS query and response.
- c) Browse to a simple website.
- d) Stop the capture tool once the page has loaded.
- e) What type filter of HTTP and DNS will be used.
- f) Filter the captured packets to isolate the DNS and HTTP traffic.
- g) After DNS resolution, explain further steps taken to establish link with web server. Elaborate using diagram.

Answer:

1. **HTTP Request/Response:**
 - Request method: **GET / HTTP/1.1**
 - Response: **HTTP/1.1 200 OK**
 - Status code: **200 (Success)**
 - Content-Type: e.g., **text/html**
2. **DNS Query & Response:**
 - After clearing DNS cache (**ipconfig /flushdns**), PC sends **DNS query** for **www.example.com**.

- DNS server replies with **A record = 93.184.216.34** + TTL value.
- 3. **Browsing & Capture:**
 - Open browser → enter website → page loads.
 - Stop Wireshark capture once website fully loaded.
- 4. **Filters in Wireshark:**
 - DNS only → **dns**
 - HTTP only → **http**
 - Combined → **dns || http**
- 5. **Traffic Isolation:**
 - Apply above filters → view only DNS queries/responses and HTTP GET/Response packets.
- 6. **Steps after DNS Resolution (Establishing Link):**
 - **TCP 3-way handshake:**
 - SYN → SYN-ACK → ACK
 - **HTTP GET request** sent by client.
 - **Server replies with HTTP Response (200 OK)** and content.



Question # 3:

Explain Traceroute, TTL and ICMP relation and behavior while you traceroute from source 192.168.10.10 to destination 145.145.145.45. explain step by step.

Answer:

Traceroute discovers the path from a source to a destination by sending probe packets with increasing **TTL** values. Each router that decrements the TTL to 0 returns an **ICMP Time Exceeded** message, revealing that hop. When a probe reaches the destination, the destination replies (type depends on probe): traceroute then stops.

Step-by-step behavior

Assume source **192.168.10.10** and destination **145.145.145.45**:

1. **TTL = 1:**

- Host sends a probe to 145.145.145.45 with TTL=1.
 - Router#1 receives it, decrements TTL → becomes 0. Router#1 discards the packet and sends **ICMP Type 11 (Time Exceeded)** back to 192.168.10.10. Traceroute records hop 1 and the round-trip time (RTT).
2. **TTL = 2:**
Host sends probes with TTL=2. Router#1 forwards (TTL→1), Router#2 decrements to 0 → Router#2 sends **ICMP Time Exceeded** back. Traceroute records hop 2.
3. **Repeat:**
TTL is increased (3, 4, ...). Each router that causes TTL→0 replies with ICMP Time Exceeded, so traceroute learns successive hop IPs and RTTs.
4. **Final hop (reaches destination):**
When TTL is large enough so the probe reaches 145.145.145.45, the destination responds differently depending on probe type:
- **UDP probe (Linux/Cisco default):** destination replies **ICMP Type 3, Code 3 (Port Unreachable)** → traceroute recognizes final hop.
 - **ICMP probe (Windows `tracert`):** destination replies **ICMP Echo Reply (Type 0)** → traceroute stops.
- Traceroute prints the destination IP and final RTTs.
5. **Three probes per hop:**
By default traceroute sends three probes per TTL, so each hop line usually shows three RTT values (e.g., 1 2 ms 3 ms 2 ms 192.168.1.1).

Commands / How to run

- **Windows:** `tracert 145.145.145.45`
(sends ICMP Echo probes)
- **Linux:** `traceroute 145.145.145.45`
(default uses UDP; use `-I` for ICMP if needed: `traceroute -I 145.145.145.45`)

Cisco IOS (from router):

`Router# traceroute 145.145.145.45`

`Router# traceroute 145.145.145.45 source 192.168.10.10`

- (Cisco typically uses UDP probes by default; interactive traceroute can change options)

Interpretation & troubleshooting

- * * * = no reply for that probe (ICMP filtered, rate-limited, or lost).
 - If hops stop responding or show unexpected IPs, check firewalls/ACLs, NAT, route asymmetry, or ICMP rate-limiting on intermediate routers.
 - Useful Cisco checks: `show ip route`, `show ip interface`, and (briefly) `debug ip icmp` (use carefully).
-